W-R-175/3D
22 December 1986
SUPERSEDING
W-R-175/3C
15 May 1973

FEDERAL SPECIFICATION SHEET

# REELS, PLASTIC, FIBERGLASS, METALLIC, AND METALLIC FLANGE PHENOLIC HUB, WITH 3-INCH CENTER HOLE

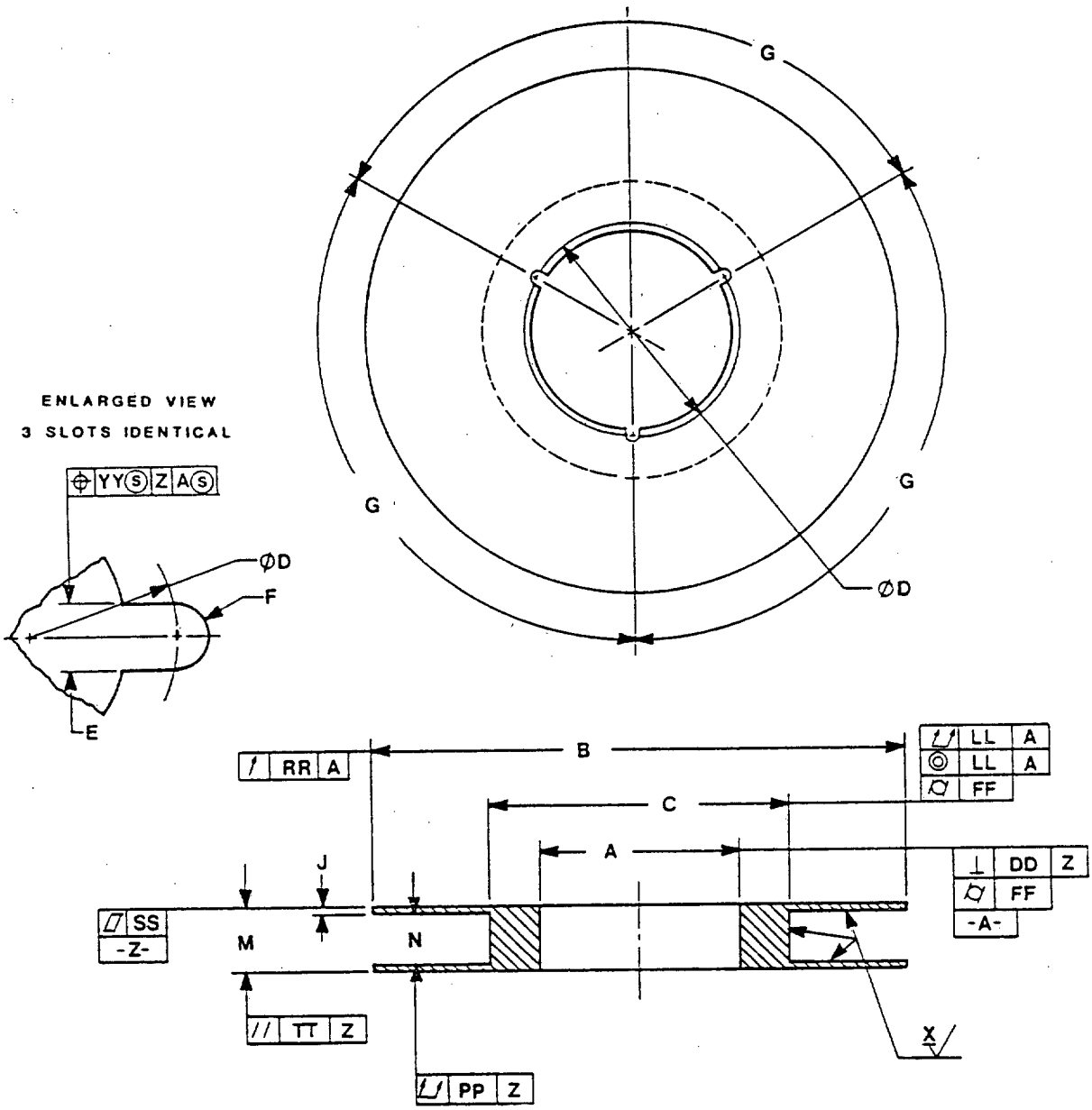
This specification sheet is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

The requirements for acquiring the reels described herein shall consist of this specification and the latest issue of W-R-175.

AMSC N/A
FSC 5835
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited



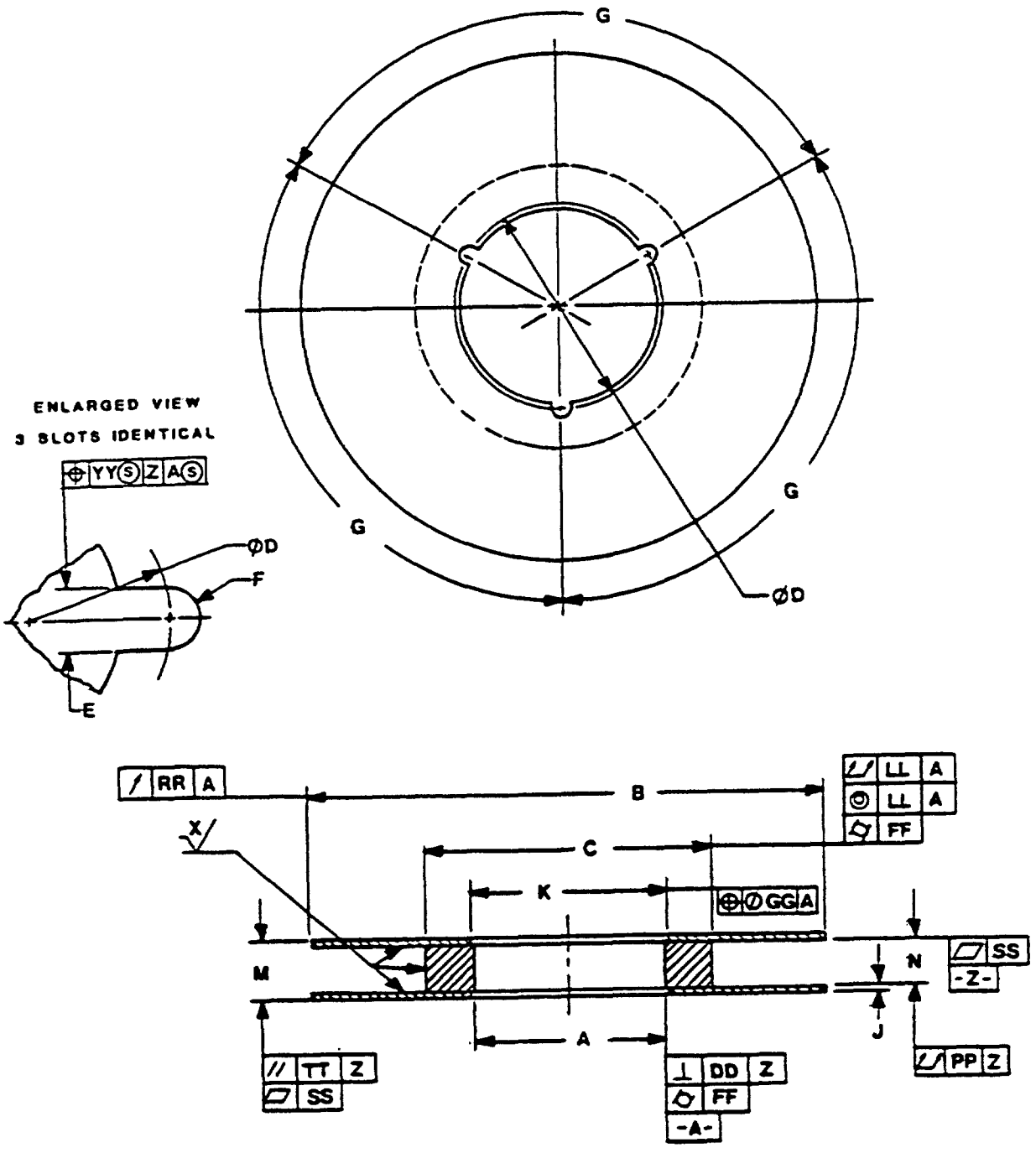
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.




SH 13202772

FIGURE 1. EB, JB and PB reel configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

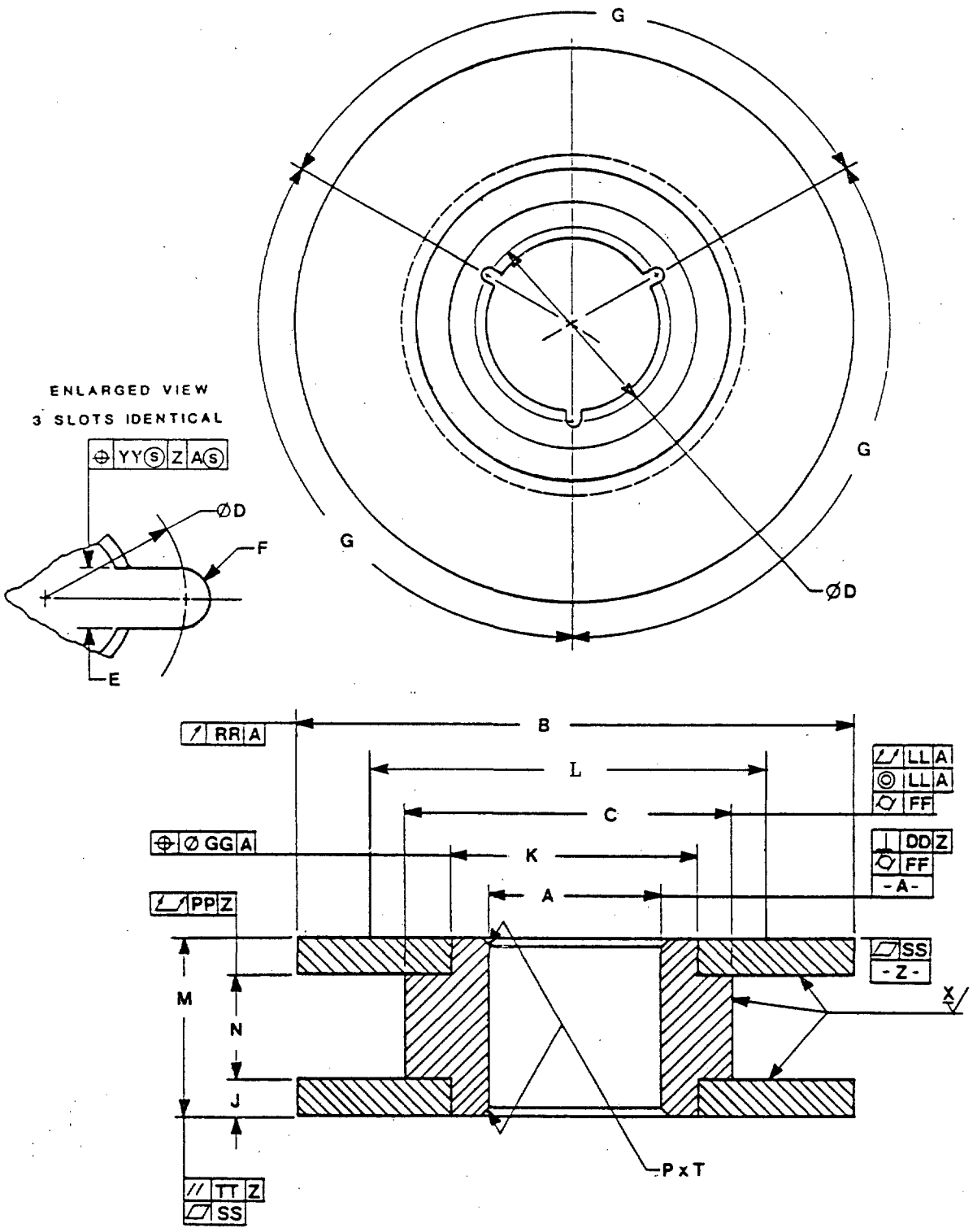



SH 13202773

FIGURE 2. EC, PC, and SC reel configuration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.




SH 13202800

FIGURE 3. ED, JD, PD, QD, and SD reel configuration.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

1. Materials:

The following reel types shall be constructed of plastic or fiberglass-filled plastic.

| Type | Nominal flange diameter |
|---|---|
| EB | 8 (203.2) |
| JB | 9.75 (247.7) |
| PB | 10.5 (266.7) |

The following type reels shall be constructed of anodized aluminum or an alloy with aluminum as the principal element:

| Type | Nominal flange diameter |
|---|---|
| EC | 8 (203.2) |
| PC | 10.5 (266.7) |
| SC | 14 (355.6) |

The following type reels shall be constructed of anodized aluminum flanges fastened to a phenolic resin hub:

| Type | Nominal flange diameter |
|---|---|
| ED | 8 (203.2) |
| JD | 9.75 (247.7) |
| PD | 10.5 (266.7) |
| QD | 12.5 (317.5) |
| SD | 14 (355.6) |

Flange fastening method shall be as follows:

Type EC, PC, and SC reels shall have flanges fastened to the hub by screws or nuts and bolts. Flange fastening members shall be flush with or below the outer surface of the flanges.

Type ED, JD, PD, QD, and SD reels shall have flanges fastened to the hub by rivets extending through the hub or by screws tapped into the hub. The fasteners shall not protrude above the lateral mounting surface.

2. Dimensions: Reel dimensions shall be as specified in tables I, II, and III (see figures 1, 2 and 3).

3. Friction rings: When specified, reels shall be furnished with friction rings.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions for EB, JB, and PB reel.[1] [2]

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Inside hub dia (A) | | 3.025 (76.835) | 3.010 (76.454) | |
| Outside flange dia (B) | EB | 8.020 (203.71) | 7.980 (202.69) | 0.030 (0.762) |
| | JB | 9.770 (248.16) | 9.730 (247.14) | .040 (1.016) |
| | PB | 10.520 (267.21) | 10.480 (266.19) | .050 (0.127) |
| Outside hub dia[4] (C) | | 4.510 (114.56)[5] | 4.490 (114.05) | |
| Keyslot pitch dia (D) | | 3.270 (83.06) | 3.248 (82.50) | |
| Keyslot width (E) | | 0.229 (5.817) | 0.219 (5.563) | |
| Keyslot radius (F) | | .5E | .5E | |
| Angle between successive keyslots (G) (degrees) | | 120.25 (2.099rad) | 119.75 (2.090rad) | |
| Flange thickness (J) | | 0.098 (2.489) | 0.085 (2.133) | |
| | Tape width indicator | | | |
| Overall reel width (M) within lateral mounting surface | 250 | 0.486 (12.34) | 0.458 (11.63) | |
| | 500 | .738 (18.75) | .710 (18.03) | |
| | 750 | .988 (25.10) | .960 (24.38) | |
| | 1000 | 1.238 (31.46) | 1.210 (30.73) | |
| | 2000 | 2.238 (56.85) | 2.210 (56.13) | |
| Distance between flanges at hub[6] (N) | 250 | 0.290 (7.37) | 0.262 (6.65) | |
| | 500 | .542 (13.77) | .514 (13.06) | |
| | 750 | .792 (20.12) | .764 (19.41) | |
| | 1000 | 1.042 (26.47) | 1.014 (25.76) | |
| | 2000 | 2.042 (51.87) | 2.014 (51.16) | |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE I. Dimensions for EB, JB and PB reels.[1] [2] - Continued

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange defection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Mounting surface parallelism (TT) | | 0.010 (0.254) | | |
| Inside hub surface perpendicularity (DD) | | .008 (0.203) in/in of width | | |
| Inside hub surface cylindricity (FF) | | .008 (0.203) in/in of width | | |
| Outside hub surface cylindricity (FF) | | .004 (0.102) in/in of width | | |
| Outside hub surface concentricity (total runout) (LL) | | .010 (0.254) | | |
| Flange rim concentricity (RR) | | .050 (1.27) | | |
| Flange wobble (PP) | EB<br>JB,PB | .025 (0.635)<br>.030 (0.762) | | |
| Keyslot position (YY) | | .010 (0.254) | | |
| Tape path clearance[7] | | | | |
| Roughness of inside flange surface (X) | | 80 microinches | | |
| Roughness of outside hub surface (X) | | 80 microinches | | |
| Open flange area (percent) | | 50 | | |
| Diameter position (GG) | | 0.002 (0.051) | | |

See footnotes at top of next page.





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] Lateral mounting surfaces extend out to "C" diameter.
[5] This dimension is exclusive of any friction ring used on the reel. If a friction ring is used it shall meet the following requirements:

(a) The material shall be neoprene of a durometer hardness of 60 ± 10 or it shall be a suitable elastomer.
(b) The outside diameter shall be concentric with the inside diameter within a tolerance of 0.020 (0.51).
(c) The inside and outside surfaces shall be cylindrical within a tolerance zone of 0.010 (0.25).
(d) The friction ring shall meet the temperature and humidity requirements of W-R-175.
(e) When mounted on the reel, the ring shall be centered between the flanges and its width shall be 0.005 to 0.008 (0.127 to 0.203) less than the applicable "N" dimension.
(f) The ring shall be flat on the hub and shall exhibit no visible buckling or puckering of the ring surface cracking or slitting when mounted on the reel.

[6] The distance between the flanges, measured at the periphery of the flanges, shall not exceed the distance between the flanges at the hub (dimension "N") by more than the following applicable value for a given reel, measured within 0.130 inch from the flange periphery:

EB - 0.040 (1.016)
JB - 0.050 (1.270)
PB - 0.050 (1.270)

[7] The tape path clearance shall be measured in accordance with W-R-175. The applicable tape path clearance shall be as follows:

| Tape width indicator | Tape path center line |
|---|---|
| 250 | 0.229 (5.817) |
| 500 | .355 (9.017) |
| 750 | .480 (12.192) |
| 1000 | .605 (15.367) |
| 2000 | 1.105 (28.067) |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE II. Dimensions for EC, PC, and SC reels.[1] [2]

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Inside hub dia (A) | | 3.008 (76.403) | 3.002 (76.250) | |
| Outside flange dia (B) | EC | 8.020 (203.71) | 7.980 (202.69) | 0.030 (0.762) |
| | PC | 10.520 (267.21) | 10.480 (266.19) | .170 (4.318) |
| | SC | 14.020 (356.11) | 13.980 (355.09) | .210 (5.334) |
| Outside hub dia[4] (C) | | 4.510 (114.56)[5] | 4.490 (114.05) | |
| Keyslot pitch dia (D) | | 3.258 (82.75) | 3.248 (82.50) | |
| Keyslot width (E) | | 0.229 (5.817) | 0.219 (5.563) | |
| Keyslot radius (F) | | .5E | .5E | |
| Angle between successive keyslots (G) (degrees) | | 120.25 (2.099rad) | 119.75 (2.090rad) | |
| Flange thickness (J) | | 0.065 (1.651) | 0.048 (1.219) | |
| Inside diameter of[4] flange (K) | | 3.041 (77.241) | 3.021 (76.733) | |
| | Tape width indicator | | | |
| Overall reel width (M) within lateral mounting surface | 250 | 0.482 (12.43) | 0.442 (11.23) | |
| | 500 | .732 (18.59) | .692 (17.58) | |
| | 1000 | 1.215 (30.861) | 1.209 (30.709) | |
| Distance between flanges at hub[6] (N) | 250 | .355 (9.017) | .345 (8.763) | |
| | 500 | .605 (15.37) | .595 (15.11) | |
| | 1000 | 1.105 (28.067) | 1.095 (27.773) | |

See footnotes at end of table.





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE II. Dimensions for EC, PC, and SC reels.[1] [2] - Continued

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Mounting surface parallelism (TT) | | 0.007 (0.178) | | |
| Mounting surface flatness (SS) | | .004 (0.102) | | |
| Inside hub surface perpendicularity (DD) | | .002 (0.051) in/in of width | | |
| Inside hub surface cylindricity (FF) | | .002 (0.051) in/in of width | | |
| Outside hub surface cylindricity (FF) | | .002 (0.051) in/in of width | | |
| Outside hub surface concentricity (total runout) (LL) | | .010 (0.254) | | |
| Flange rim concentricity (RR) | | .050 (1.27) | | |
| Flange wobble (PP) | EC | .028 (0.711) | | |
| | PC | .040 (1.016) | | |
| | SC | .050 (1.270) | | |
| Keyslot position (YY) | | .010 (0.254) | | |
| Tape path clearance[7] | | | | |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE II. Dimensions for EC, PC, and SC reels.[1] [2] - Continued

| Characteristic[3] | Dimensions | | Flange deflection limit |
|---|---|---|---|
| | maximum | minimum | |
| Roughness of inside (X) flange surface | 80 microinches | | |
| Roughness of outside hub surface | 80 microinches | | |
| Open flange area (percent) | 50 | | |
| Diameter position (GG) | 0.002 (0.051) | | |

See footnotes at top of next page.





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] Lateral mounting surfaces extend out to "C" diameter.
[5] This dimension is exclusive of any friction ring used on the reel. If a friction ring is used it shall meet the following requirements:

(a) The material shall be neoprene of a durometer hardness of 60 ± 10 or it shall be a suitable elastomer.
(b) The outside diameter shall be concentric with the inside diameter within a tolerance of 0.020 (0.51).
(c) The inside and outside surfaces shall be cylindrical within a tolerance zone of 0.010 (0.25).
(d) The friction ring shall meet the temperature and humidity requirements of W-R-175.
(e) When mounted on the reel, the ring shall be centered between the flanges and its width shall be 0.005 to 0.008 (0.127 to 0.203) less than the applicable "N" dimension.
(f) The ring shall be flat on the hub and shall exhibit no visible buckling or puckering of the ring surface cracking or slitting when mounted on the reel.

[6] The distance between the flanges, measured at the periphery of the flanges, shall not exceed the distance between the flanges at the hub (dimension "N") by more than the following applicable value for a given reel, measured within 0.130 inch from the flange periphery:

EC - 0.026 (0.660)
PC - 0.050 (1.270)
SC - 0.050 (1.270)

[7] The tape path clearance shall be measured in accordance with W-R-175. The applicable tape path clearance shall be as follows:

| Tape width indicator | Tape path center line |
|---|---|
| 250 | 0.229 (5.817) |
| 500 | .355 (9.017) |
| 750 | .480 (12.192) |
| 1000 | .605 (15.367) |
| 2000 | 1.105 (28.067) |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE III. Dimensions for ED, JD, PD, QD and SD reels.[1] [2]

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Inside hub dia (A) | | 3.011 (76.479) | 3.000 (76.200) | |
| Outside flange dia (B) | ED | 8.020 (203.71) | 7.980 (202.69) | 0.030 (0.762) |
| | JD | 9.770 (248.16) | 9.730 (247.14) | .040 (1.016) |
| | PD | 10.520 (267.21) | 10.480 (266.19) | .050 (1.270) |
| | QD | 12.520 (318.01) | 12.480 (316.99) | .050 (1.270) |
| | SD | 14.020 (356.11) | 13.980 (355.09) | .080 (2.032) |
| Outside hub dia (C) | | 4.510 (114.56)[4] | 4.490 (114.05) | |
| Keyslot pitch dia (D) | | 3.258 (82.752) | 3.248 (82.499) | |
| Keyslot width (E) | | 0.225 (5.715) | 0.219 (5.563) | |
| Keyslot radius (F) | | .5E | .5E | |
| Angle between successive keyslots (degrees) (G) | | 120.25 (2.099rad) | 119.75 (2.090rad) | |
| Flange thickness (J) | | 0.065 (1.651) | 0.048 (1.219) | |
| Inside diameter of flange and mating hub diameter[5] (K) | | 3.700 (93.980)[6] | 3.600 (91.440)[6] | |
| Flange clearance between lateral mounting surfaces, dia[9] (L) | | | 6.0 (152.4) | |

See footnotes at end of table.





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE III. Dimensions for ED, JD, PD, QD and SD reels.[1] [2] - Continued

| Characteristic[3] | Tape width indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Overall reel width within lateral mounting surface (M) | 250 | 0.472 (11.99) | 0.459 (11.66) | |
| | 500 | .722 (18.34) | .709 (18.00) | |
| | 750 | .972 (24.69) | .959 (24.36) | |
| | 1000 | 1.222 (31.04) | 1.209 (30.71) | |
| | 2000 | 2.222 (56.44) | 2.209 (56.11) | |
| Distance between flanges at hub[7] (N) | 250 | 0.292 (7.42) | 0.279 (7.09) | |
| | 500 | .542 (13.77) | .529 (13.44) | |
| | 750 | .792 (20.12) | .779 (19.79) | |
| | 1000 | 1.042 (26.47) | 1.029 (26.14) | |
| | 2000 | 2.027 (51.49) | 2.014 (51.16) | |
| Mounting surface parallelism (TT) | | 0.003 (0.076) | | |
| Mounting surface flatness (SS) | | .002 (0.051) | | |
| Inside hub surface perpendicularity (DD) | | .004 (0.102) in/in of width | | |
| Inside hub surface cylindricity (FF) | | .004 (0.102) in/in of width | | |
| Outside hub surface cylindricity (FF) | | .002 (0.051) | | |
| Outside hub surface concentricity (total runout) (LL) | | .005 (0.127) | | |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

TABLE III. Dimensions for ED, JD, PD, QD and SD reels.[1] [2] - Continued

| Characteristic[3] | Reel and hub indicator | Dimensions | | Flange deflection limit |
|---|---|---|---|---|
| | | maximum | minimum | |
| Flange rim concentricity (RR) | | 0.020 (0.51) | | |
| Flange wobble (PP) | ED | .018 (0.457) | | |
| | JD | .030 (0.762) | | |
| | PD | .023 (0.584) | | |
| | QD | .026 (0.660) | | |
| | SD | .030 (0.762) | | |
| Keyslot position (YY) | | .010 (0.254) | | |
| Tape path clearance[8] | | | | |
| Roughness of inside flange surface (X) | | 50 microinches | | |
| Roughness of outside hub surface (X) | | 50 microinches | | |
| Open flange area (percent) | | 25 | | |
| Hub chamfer angle (deg) (P) | | 46.00 (0.80rad) | 44.00 (0.77rad) | |
| Hub chamfer (T) | | 0.070 (1.78) | 0.050 (1.27) | |
| Diameter position (GG) | | .002 (0.051) | | |

See footnotes at top of next page.





Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

[1] Unless otherwise specified, all measurements are in inches. Metric equivalents are given for information only.
[2] All sharp edges shall be rounded to at least 0.004 (0.102) radius.
[3] Where applicable, tolerance zones are indicated in parentheses.
[4] This dimension is exclusive of any friction ring used on the reel. If a friction ring is used it shall meet the following requirements:

(a) The material shall be neoprene of a durometer hardness of 60 ± 10 or it shall be a suitable elastomer.
(b) The outside diameter shall be concentric with the inside diameter within a tolerance of 0.020 (0.51).
(c) The inside and outside surfaces shall be cylindrical within a tolerance zone of 0.010 (0.25).
(d) The friction ring shall meet the temperature and humidity requirements of W-R-175.
(e) When mounted on the reel, the ring shall be centered between the flanges and its width shall be 0.005 to 0.008 (0.127 to 0.203) less than the applicable "N" dimension.
(f) The ring shall be flat on the hub and shall exhibit no visible buckling or puckering of the ring surface cracking or slitting when mounted on the reel.

[5] Lateral mounting surfaces extend out to "K" diameter.
[6] The "K" diameter for the flange shall be 3.700 maximum, and the "K" diameter for the hub shall be 3.600 minimum. The "K" diameter is included for reference purposes only. The "K diameter will not be tested.
[7] The distance between the flanges, measured at the periphery of the flanges, shall not exceed the distance between the flanges at the hub (dimension "N") by more than the following applicable value for a given reel, measured within 0.130 inch from the flange periphery:

ED - 0.026 (0.660)
JD - 0.050 (1.270)
PD - 0.036 (0.914)
QD - 0.042 (1.067)
SD - 0.050 (1.270)

[8] The tape path clearance shall be measured in accordance with W-R-175. The applicable tape path clearance shall be as follows:

| Tape width indicator | Tape path center line |
|---|---|
| 250 | 0.229 (5.817) |
| 500 | .355 (9.017) |
| 750 | .480 (12.192) |
| 1000 | .605 (15.367) |
| 2000 | 1.105 (28.067) |

[9] "L" is the minimum diameter within which the flange, its fasteners and/or any labeling shall not protrude outside the planes of the lateral mounting surfaces.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

4. Mounting: When specified, "EB", "JB", "PB", "ED", "JD", "PD", "QD" and "SD" may be supplied such that reel mounts from only one side.

QUALIFICATION:

1. The qualification sample for "EB", "JB", and "PB" reels shall consist of ER-1000-JB reels. The qualification sample for "EC", "PC", and "SC" reels shall consist of ER-500-PC reels. The qualification sample for "ED", "JD", and "PD" reels shall consist of ER-1000-JD reels. The qualification sample for "QD" and "SD" reels shall consist of ER-2000-QD reels.

2. Qualification of ER-1000-JB reels shall constitute qualification for all "EB", "JB", and "PB" reels, of the configuration tested. Qualification of ER-500-PC reels shall constitute qualification for all "EC", "PC", and "SC" reels regardless of reel width. Qualification of ER-1000-JD reels shall constitute qualification for all "ED", "JD", and "PD" reels, regardless of reel width. Qualification of ER-2000-QD reels shall constitute qualification for all "QD" and "SD" reels regardless of reel width.

3. Completely assembled "EB", "EC", "JB", "PB", "PC", and "SC" reels shall be measured for mounting surface parallelism and outside hub surface cylindricity as specified in W-R-175.

MILITARY INTERESTS:

Custodians

Army - CR
Navy - SH
Air Force - 99

Review activities

Navy - NV, OM
Air Force - 80
DLA - ES

User activity

Navy - AS

CIVIL COORDINATING ACTIVITY:
NASA - NPPO

Preparing activity:
Navy - SH
(Project 5835-0079-3)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.